

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**  
**Applied Business Analytics**

**Practical – 1**

|                              |                                   |
|------------------------------|-----------------------------------|
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| Class: TYIT                  | Batch: 1                          |
| Date of Assignment: 24/07/26 | Date/Time of Submission: 24/07/26 |

a. Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset.

**Code & Output:**

Step 1: imports all necessary files.

Step 2: load the data set and display the first & last 5 rows.

Step 3: display the shape of dataset and column name & information.

Step 4: Display data type and check the missing value & duplicate records.

Step 5: Display descript static, count uniq value & display unique value of particular columns

```
#Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#Load dataset
df = pd.read_csv("Retail_sales.csv")
```

```
#Display First 5 Records
df.head()
```

|   | Store ID   | Product ID | Date       | Units Sold | Sales Revenue (USD) | Discount |
|---|------------|------------|------------|------------|---------------------|----------|
| 0 | Spearsland | 52372247   | 2022-01-01 | 9          | 2741.69             |          |
| 1 | Spearsland | 52372247   | 2022-01-02 | 7          | 2665.53             |          |
| 2 | Spearsland | 52372247   | 2022-01-03 | 1          | 380.79              |          |
| 3 | Spearsland | 52372247   | 2022-01-04 | 4          | 1523.16             |          |
| 4 | Spearsland | 52372247   | 2022-01-05 | 2          | 761.58              |          |

```
#Display Last 5 Records
df.tail()
```

|       | Store ID   | Product ID | Date       | Units Sold | Sales Revenue (USD) | Disco |
|-------|------------|------------|------------|------------|---------------------|-------|
| 29995 | Spearsland | 50239115   | 2022-01-25 | 5          | 2501.15             |       |
| 29996 | Spearsland | 50239115   | 2022-01-26 | 3          | 1500.69             |       |
| 29997 | Spearsland | 50239115   | 2022-01-27 | 6          | 3001.38             |       |
| 29998 | Spearsland | 50239115   | 2022-01-28 | 5          | 2501.15             |       |
| 29999 | Spearsland | 50239115   | 2022-01-29 | 3          | 1425.66             |       |

```
#Display Shape of Dataset
df.shape
```

```
(30000, 11)
```

```
#Display Column Names
df.columns
```

```
Index(['Store ID', 'Product ID', 'Date', 'Units Sold', 'Sales Revenue (USD)',
       'Discount Percentage', 'Marketing Spend (USD)', 'Store Location',
       'Product Category', 'Day of the Week', 'Holiday Effect'],
      dtype='object')
```

```
#Display Dataset Information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Store ID               30000 non-null  object
1   Product ID             30000 non-null  int64
2   Date                   30000 non-null  object
3   Units Sold             30000 non-null  int64
4   Sales Revenue (USD)    30000 non-null  float64
5   Discount Percentage     30000 non-null  int64
6   Marketing Spend (USD)  30000 non-null  int64
7   Store Location          30000 non-null  object
8   Product Category       30000 non-null  object
9   Day of the Week        30000 non-null  object
10  Holiday Effect          30000 non-null  bool
dtypes: bool(1), float64(1), int64(4), object(5)
memory usage: 2.3+ MB
```

```
#Display Data Types
df.dtypes
```

|  | Store ID | Product ID | Date   | Units Sold | Sales Revenue (USD) | Discount Percentage | Marketing Spend (USD) | Store Location | Product Category | Day of the Week | Holiday Effect |
|--|----------|------------|--------|------------|---------------------|---------------------|-----------------------|----------------|------------------|-----------------|----------------|
|  | object   | int64      | object | int64      | float64             | int64               | int64                 | object         | object           | object          | bool           |

```
#Display Descriptive Statistics
df.describe()
```

|       | Product ID   | Units Sold   | Sales Revenue (USD) | Discount Percentage | Marketing Spend (USD) |
|-------|--------------|--------------|---------------------|---------------------|-----------------------|
| count | 3.000000e+04 | 30000.000000 | 30000.000000        | 30000.000000        | 30000.000000          |
| mean  | 4.461294e+07 | 6.161967     | 2749.509593         | 2.973833            | 49.944033             |
| std   | 2.779759e+07 | 3.323929     | 2568.639288         | 5.974530            | 64.401655             |
| min   | 3.636541e+06 | 0.000000     | 0.000000            | 0.000000            | 0.000000              |
| 25%   | 2.228600e+07 | 4.000000     | 882.592500          | 0.000000            | 0.000000              |
| 50%   | 4.002449e+07 | 6.000000     | 1902.420000         | 0.000000            | 1.000000              |
| 75%   | 6.559352e+07 | 8.000000     | 3863.920000         | 0.000000            | 100.000000            |
| max   | 9.628253e+07 | 56.000000    | 27165.880000        | 20.000000           | 199.000000            |

```

#Check Missing Values
df.isnull().sum()

Store ID      0
Product ID    0
Date          0
Units Sold    0
Sales Revenue (USD) 0
Discount Percentage 0
Marketing Spend (USD) 0
Store Location 0
Product Category 0
Day of the Week 0
Holiday Effect 0
dtype: int64

#Check Duplicate Records
df.duplicated().sum()

np.int64(0)

#Count Unique Values
df.nunique()

Store ID      1
Product ID    42
Date          731
Units Sold    32
Sales Revenue (USD) 2545
Discount Percentage 5
Marketing Spend (USD) 200
Store Location 243
Product Category 4
Day of the Week 7
Holiday Effect 2
dtype: int64

#Display Unique Values of a Particular column
df["Product Category"].unique()

array(['Furniture', 'Electronics', 'Groceries', 'Clothing'], dtype=object)

```

Step 6: count product category and count store location, holliday effect.

Step 7: calculate sale revenue , maximum & minimum sale revenue , correlation matrix.

```

#Count Product Categories
df["Product Category"].value_counts()

Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64

#Count Store Locations
df["Store Location"].value_counts()

Store Location
Korea          237
Congo          234
Turks and Caicos Islands 156
Guam           153
Tanzania       151
...
Trinidad and Tobago 98
Haiti           96
Dominica        95
Ireland         92
Andorra         80
Name: count, Length: 243, dtype: int64

#Count Holiday Effect
df["Holiday Effect"].value_counts()

Holiday Effect
False      29836
True       164
Name: count, dtype: int64

#Calculate Average Sales Revenue
df["Sales Revenue (USD)"].mean()

np.float64(2749.509592666667)

#Find Maximum Sales Revenue
df["Sales Revenue (USD)"].max()

np.float64(27165.88)

#Find Minimum Sales Revenue
df["Sales Revenue (USD)"].min()

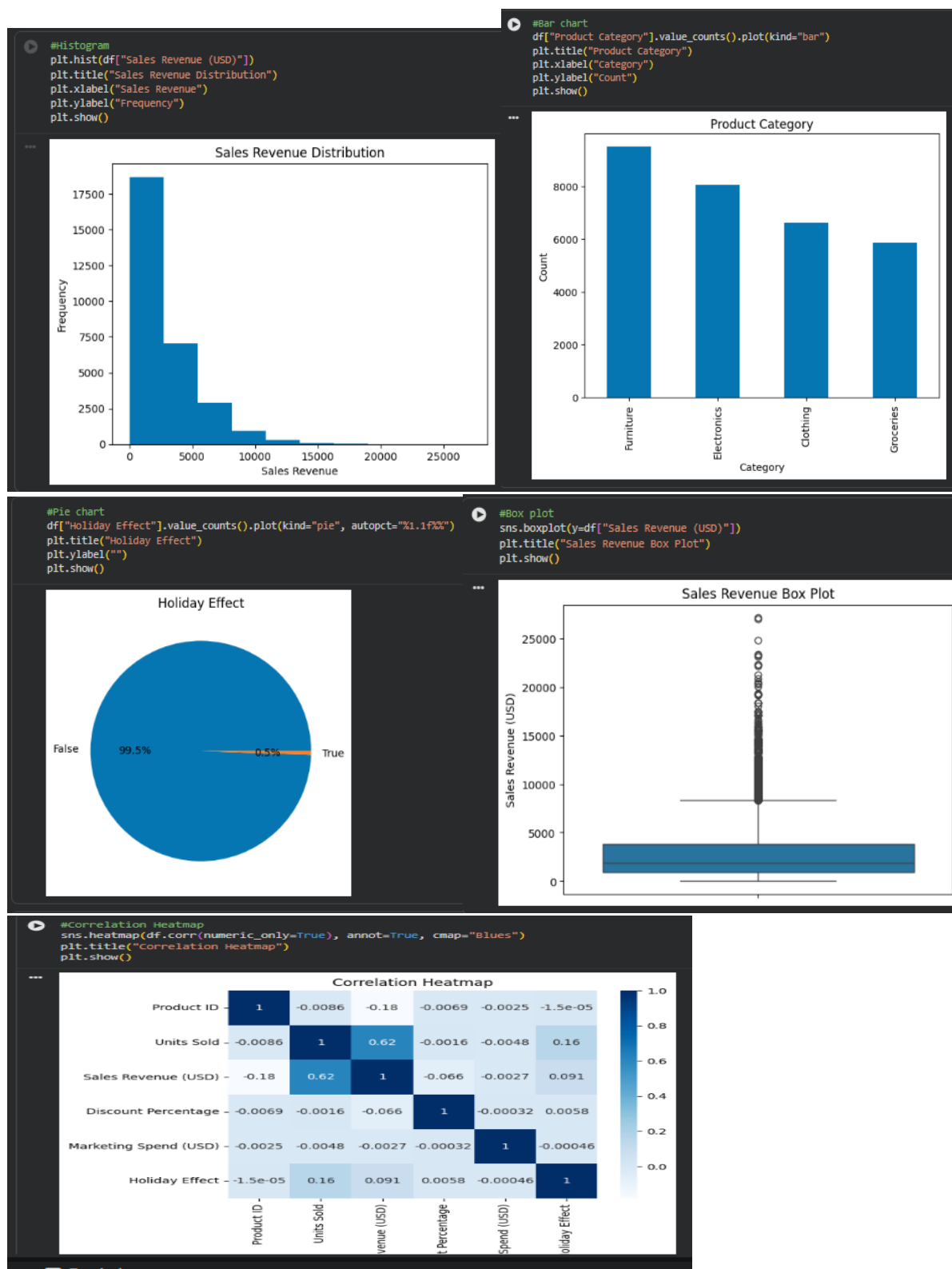
np.float64(0.0)

#Correlation Matrix
df.corr(numeric_only=True)

Product ID  Units Sold  Sales Revenue (USD)  Discount Percentage  Marketing Spend (USD)  Holiday Effect
Product ID      1.000000      -0.008570        -0.181934        -0.006891        -0.002496      -0.000015
Units Sold      -0.008570      1.000000         0.615767        -0.001620        -0.004750       0.159324
Sales Revenue (USD) -0.181934      0.615767         1.000000        -0.065791        -0.002732       0.090681
Discount Percentage -0.006891     -0.001620        -0.065791         1.000000        -0.000322       0.005848
Marketing Spend (USD) -0.002496     -0.004750        -0.002732        -0.000322         1.000000      -0.000455
Holiday Effect    -0.000015     0.159324         0.090681         0.005848        -0.000455       1.000000

```

Step 8: Make a histogram , Bar, pie , Box and correlation heatmap.



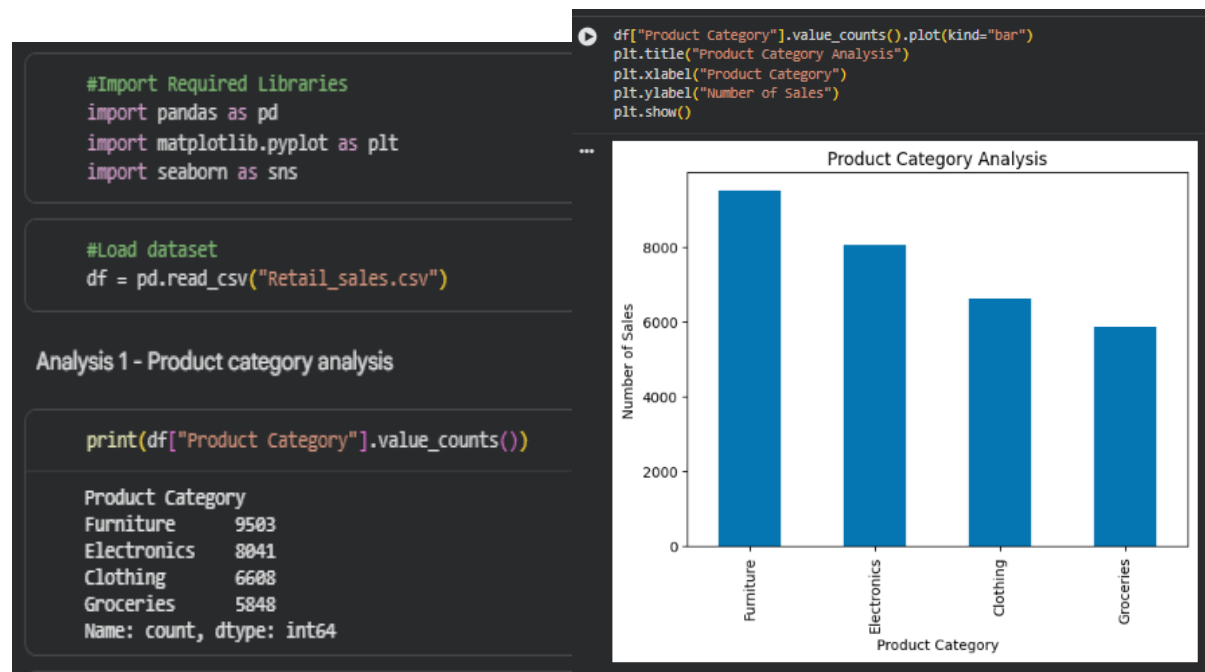
b. Analyze customer data to identify trends and patterns that can be used for business decision-making.

**Code & Output:**

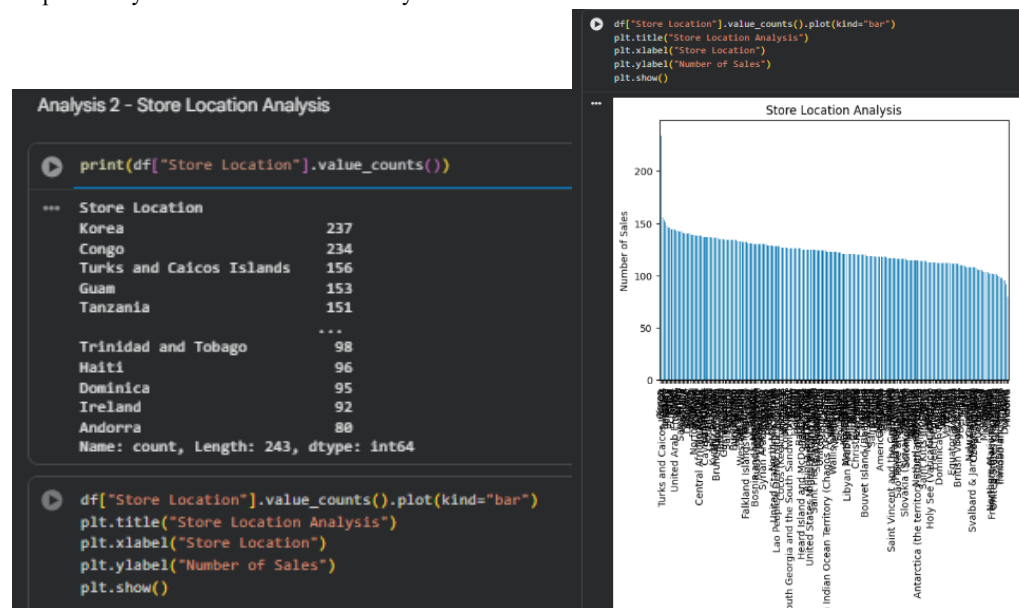
Step 1: Import Required Libraries

Step 2: load the dataset

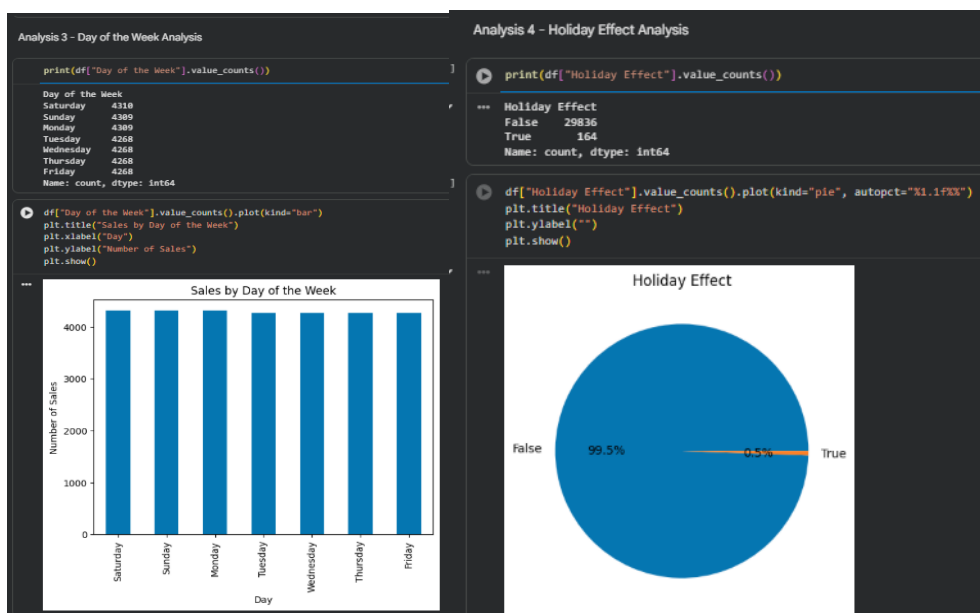
Step 3: Analysis 1 - Product category analysis



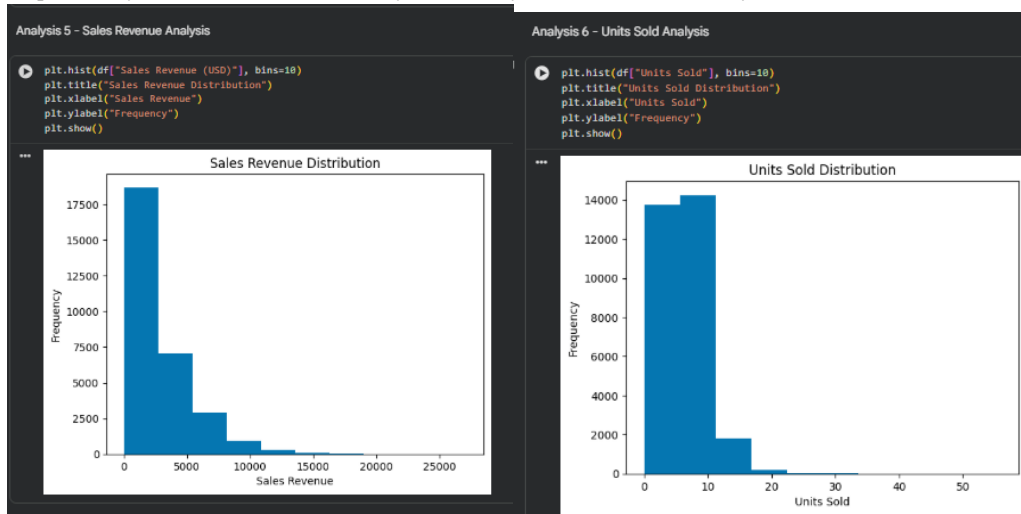
Step 4: Analysis 2 - Store Location Analysis



Step 5: Analysis 3 - Day of the Week Analysis & Analysis 4 - Holiday Effect Analysis



Step 6: Analysis 5 - Sales Revenue Analysis &amp; Analysis 6 - Units Sold Analysis



Step 7: Analysis 7 - Marketing Spend vs Sales Revenue &amp; Analysis 8 - Correlation Analysis

